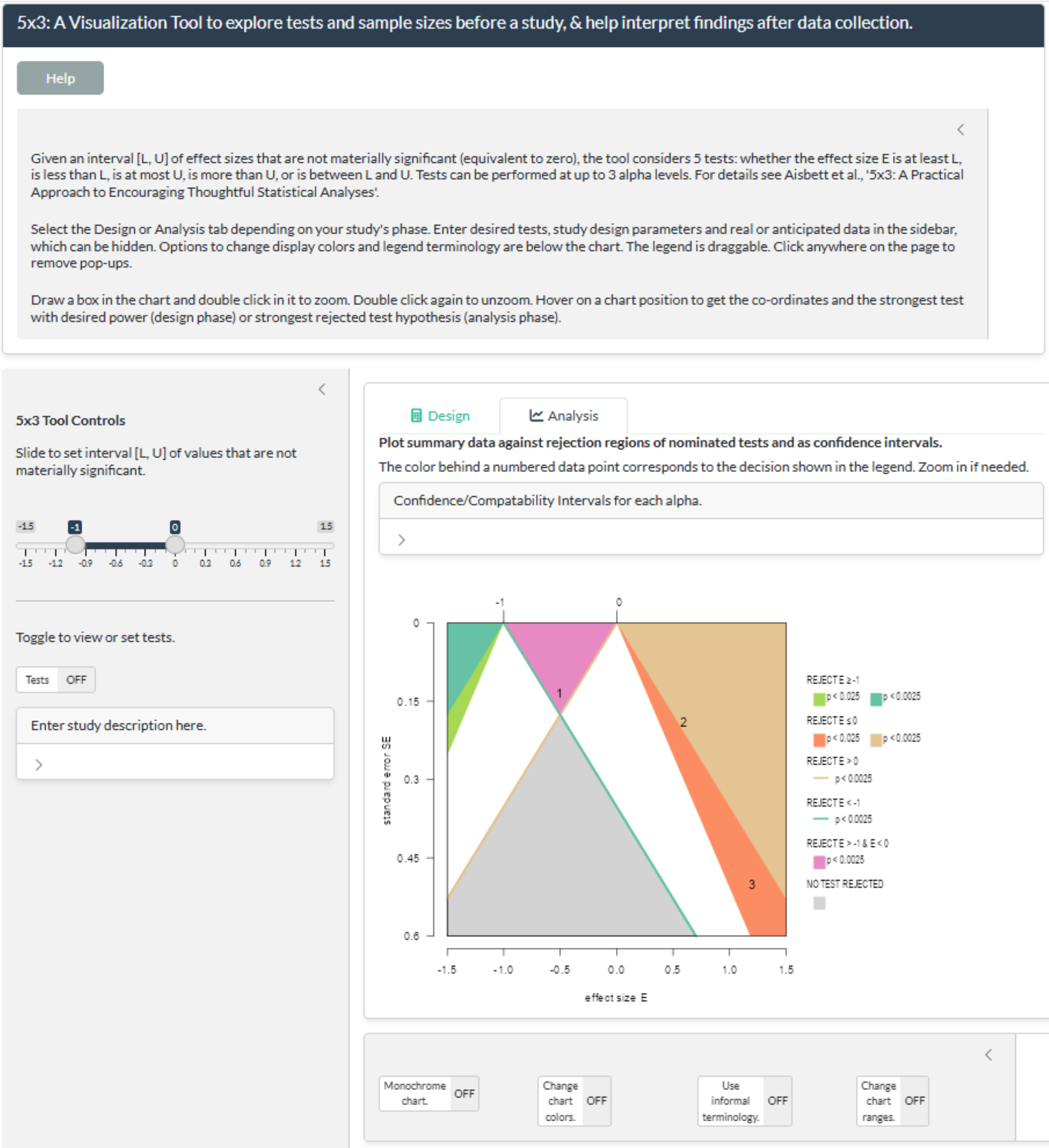
User Manual for the 5x3 App

1 OVERVIEW

5x3 is a tool for statistical design and analyses using t-tests, available as a web application at meraglimja.shinyapps.io/5x3Tool. The accompanying document, 5x3: *A Practical Approach to Encouraging Thoughtful Statistical Analyses*, explains how the theory behind the tool justifies its simultaneous testing of multiple hypotheses about the same parameter value, at multiple alpha levels. The accompanying *Handbook on the Use of 5x3 in Collaborative Statistical Design and Reporting* provides guidance on how to use the tool in collaborating with stakeholders. These documents can be accessed through the Help button at the top of the display.



The opening display is shown above. There are 4 panes:

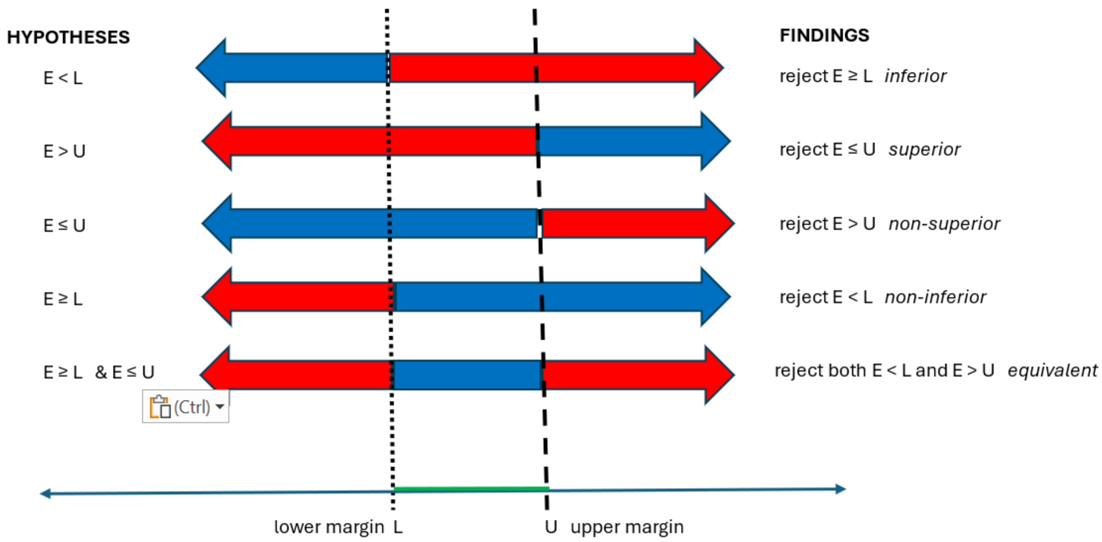
1. The top pane briefly describes 5 x 3 testing and summarises key functionality of the app.
2. To the left is a control pane with a slider sets the interval of effects that are not materially significant. Desired tests and study-related parameters and data are also specified here.
3. The main chart pane allows a choice between Design and Analysis tabs, with black and green lettering respectively indicating the active and inactive tab. The Design chart allows required sample sizes for multiple tests to be read off, given an anticipated sample size. When findings are plotted on an Analysis chart, the tests that are satisfied at various alphas can be read off. The legend that links colored regions and lines to tests is draggable within the main pane.
4. The bottom pane controls options for the look of the chart and the contents of the draggable legend.

All panes other than the chart display can be closed. The charts can be zoomed, and hovering on the chart opens pop-ups containing point co-ordinates and related information.

2 THE 5 HYPOTHESES AND 5x3 DEFAULT TERMINOLOGY FOR TESTS

Hypotheses and tests in this manual are described using the terminology commonly used in clinical trial findings. The tool by default uses more formal terminology which can be changed through the ***Use informal terminology*** switch (see 4.5.3).

The core idea is that, instead of testing for no effect, you conduct one-sided tests against the margins of the interval of effect sizes considered to be not materially or clinically relevant, shown as the black range on the slider in the earlier screenshot and as broken lines in the graphic below.



The margins of this interval determine 5 hypotheses listed to the left in the graphic below, which are satisfied by effect sizes in the ranges indicated by the blue arrows. The *test hypotheses* that we seek to reject are that the effect size is in the range indicated by the *red* arrow.

The terms *inferior, superior* etc. in italics to the right of the graphic are often used in reporting findings from clinical trials, rather than the formal test result shown alongside. In clinical trials, the effect size is typically the difference between measurements in treatment and control groups, and the upper margin U is generally set to zero. A treatment is called *inferior* if the mean difference is less than the lower margin and the test E ≥ L is rejected at the specified alpha level. It is *superior* if it greater than the upper margin and the test E ≤ U is rejected at a specified level. A treatment is *non-superior* if the difference is less than the upper margin and the test E > U is rejected, while it is *non-inferior* if it is greater than the lower margin and the test E < L is rejected. The new treatment is practically *equivalen*t to the control if the effect size lies in the interval between the two margins and the tests E < L and E > U are both rejected.

The literal description of findings, such as "reject E < L", are the default in the 5x3 tool.

3 THE CHARTS

The horizontal axis of the chart represents anticipated effect size, with numbered whiskers at the top of the chart indicating the bounds L and U. The vertical axis represents standard error when the Analysis tab is active, and total sample size when the Design tab is active: note that the sample size scale isn't linear.

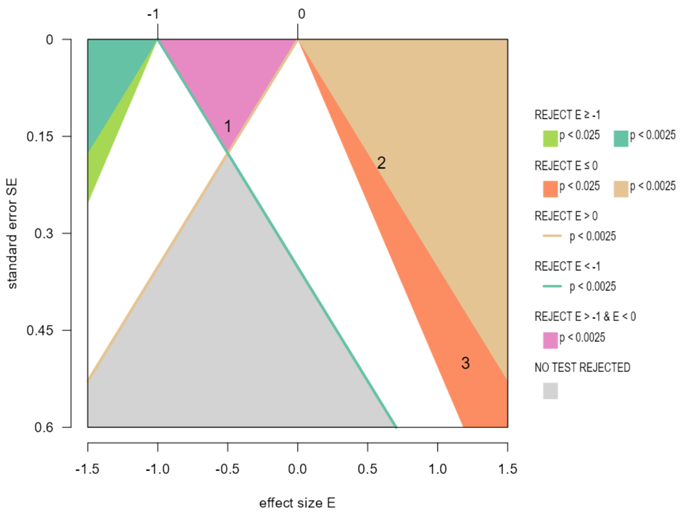
The legend maps colored regions and lines to tests at specified alpha levels. The legend can be dragged anywhere on the display pane.

Zoom in on the chart by drawing a box then double clicking. Repeat to zoom further in. Double click once or twice without drawing a box to go back. (The zoom doesn't work on mobile devices unfortunately, but you can reset the chart ranges as described below.)

A pop-up appears when you hover over a point, giving the location and other information. Click anywhere off the pane to close it.

3.1 Analysis phase

The Analysis phase chart below shows effect size on the horizontal axis and standard error on the vertical axis. The whiskers at the top of the chart show that here the margins interval equivalent to zero are L = -1 and U = 0.



The teal green region in the upper left consists of points (potential findings) with effect size less than L and whose standard error is small enough that the hypothesis E ≥ L is expected to be rejected in at least .9975 of repeated trials. This fraction falls to .975 in the bright green region. In standard terminology, the p values of findings that fall in the teal region are at most 0.0025, while findings that fall in the bright green region have p values at most 0.025. This is shown in the legend, which also specifies, for example, that findings that fall in the mauve-pink region would have a p value at most 0.0025 for the equivalence test E < L & E > U.

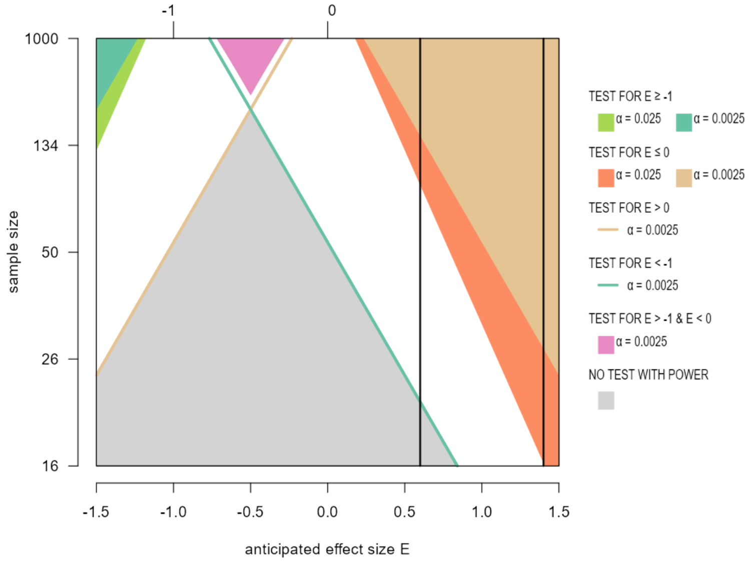
Findings that fall *above* the beige line emanating from the whisker at U = 0 are where the test E > U is expected to be rejected in at least 0.9975 of trials – that is where the p value for this test is at most 0.0025. The teal line is interpreted analogously as defining the boundary of the rejection region for the test E < L.

Actual study findings are entered in the side panel and displayed as numbers. Their interpretation is given by reading from the legend: thus, the Study 1 finding might be reported as equivalence at level 0.0025, while the Study 3 finding is of superiority at level 0.025. Section 5 discusses this further.

3.2 Design phase

In the Design phase, the chart points are pairs of anticipated effect size and total sample size, and the regions and lines define where the various tests have the nominated power to correctly reject the test at that alpha level. In this example, all tests were required to have power 80%.

For example, in the screenshot below, the orange and beige regions are all the points where superiority tests would have the required power to detect that anticipated effect size at that sample size. The legend explains that the beige region corresponds to points with the required power at alpha level 0.0025 while points in the orange region only have the required power at alpha 0.005. Similarly with the teal and bright green regions and inferiority tests.

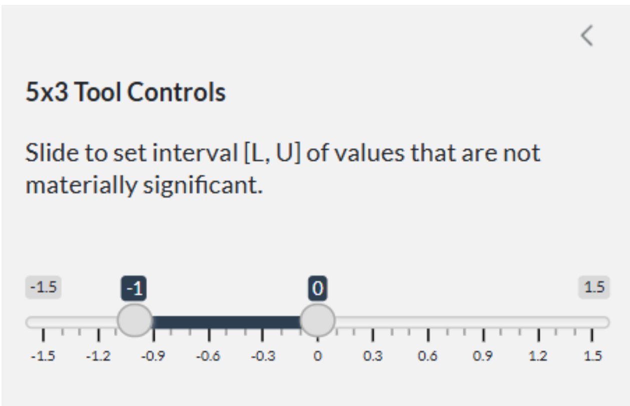


Points at which tests for non-superiority (and non-inferiority) have the nominated power are above the colored lines described in the legend. In the screenshot, points above the teal line have this power for a test at alpha level 0.0025.

The vertical lines in the chart are at anticipated effect sizes 0.6 and 1.4 (for example, stakeholders might disagree about the likely size, or more than one intervention is to be tested on the one set of subjects). The minimum sample size to attain the desired power for a test is read from where the vertical line enters the region corresponding to that test. Section 6 describes this in detail.

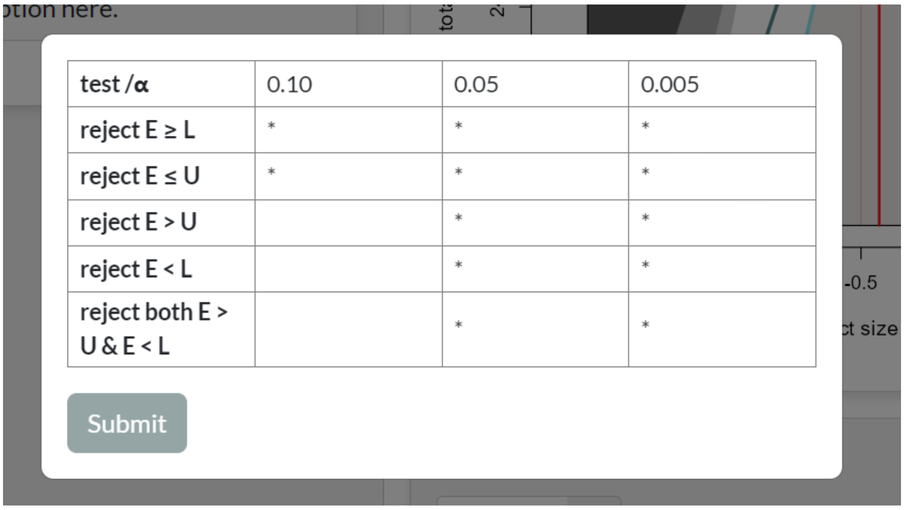
4 5x3 TOOL CONTROLS

4.1 Setting the bounds L and U



The slider at the top of the Control pane is used to set the interval of effect sizes that are not materially significant (i.e. that are equivalent to zero). The range of the slider corresponds to the range of effect sizes displayed in the chart, which can be changed (see 4.5.4). Changing the range doesn't affect the slider settings.

4.2 Setting the tests to be conducted and their alpha level(s)



test /α	0.10	0.05	0.005
reject E ≥ L	*	*	*
reject E ≤ U	*	*	*
reject E > U		*	*
reject E < L		*	*
reject both E > U & E < L		*	*

The tests to be conducted can be viewed or changed with the switch beneath the slider. This switch opens a pop-up containing a table and a button to submit changes. Click anywhere off the pop-up to get rid of it.

The table rows are automatically labelled with terminology to report rejection of the five test hypotheses: either the defaults, as shown above, or your preferred terminology (see 4.5.3).

Enter from one to three alpha levels in the top row. After you submit a revised table, the levels will be placed in decreasing order (increasing stringency of the tests) and any columns headed by non-numeric entries or entries not within the interval (0, .5) will be filled with blanks.

Put an asterisk in cells corresponding to any hypothesis you want tested at the corresponding alpha level. For example, the table in the screenshot above will result in simultaneous testing of all the hypotheses at levels 0.05 and 0.005, but will only test the hypotheses E ≥ L and E ≤ U at the weak level 0.10

Enter anything other than an asterisk into the table and it will be automatically converted to a blank. Any asterisk in the bottom row will also be converted to a blank unless both the component tests are selected at that alpha level. In clinical trial terminology, this is because equivalence implies that the new treatment is both non-inferior and non-superior to the control.

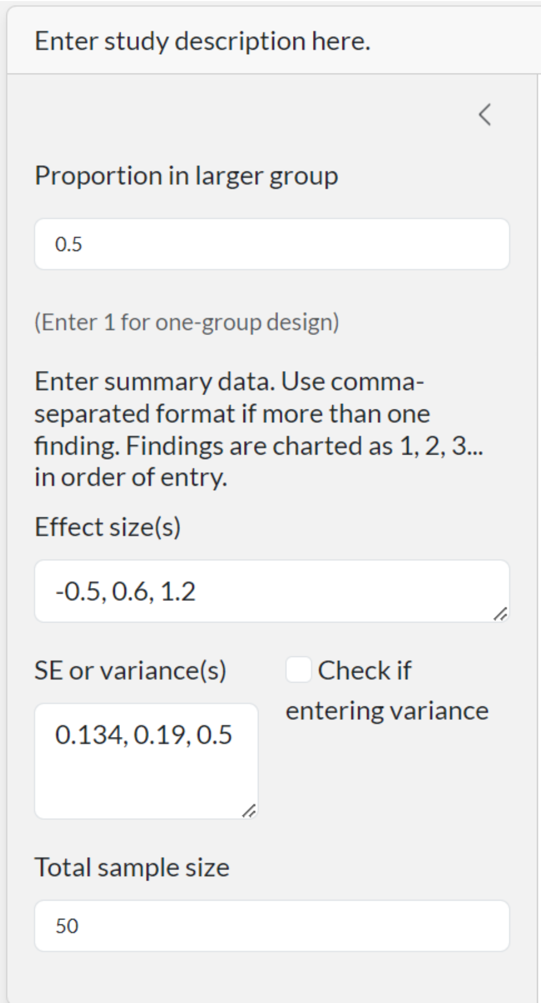
4.3 Setting power in the Design phase

When the Design tab is active, a switch is also provided to enter desired power (as a percentage) for each of the tests selected in the test table. In general, you would require lower power for a test at a higher alpha level, say 0.005, than for one at 0.05. And because rejecting E < L, say, is less specific than rejecting E ≤ U, you might require the first test to have more power at the same alpha level.

It doesn't matter what you put in cells that don't correspond to tests recorded in the top table, provided it is a number between 0 and 100. Anything else will converted to zero.

4.4 Setting parameters of your study

Currently the tool only caters for t-tests with single or two-group designs. Open the box at the bottom of the side panel to specify the design, by entering 1 for single group or entering the size of the larger group.



The graphic shows the ***Study Description*** box open with the Analysis tab selected. You must enter your calculated mean effect size and standard error (or, optionally, variance). If you want to show summary data from more than one experimental finding, enter these as comma-separated lists of effect sizes and standard errors (variances). The findings will be shown as numbered points on the chart, as described earlier.

If the Design tab is active, you will be asked to supply the anticipated effect size and sample variance, which are required in any sample size calculation. Estimating these values is notoriously difficult— there may be good arguments to support different values. Therefore the 5x3 tool allows you to enter multiple anticipated effect sizes, as a comma separated list. If you want to look at different variances though, you have to do them one at a time as you can't enter a list. Currently, in two-group designs the group variances must be same.

Finally, in both tabs, total sample size must be entered. In the Design phase, this is of course a preliminary estimate. Standard sample size calculators use normal approximations to t-tests, so you could just set this to a large value, and possibly refine it later.

4.5 5x3 Display switches

The four switches beneath the chart allow you to alter the chart’s appearance.

4.5.1 Switching to a monochrome display

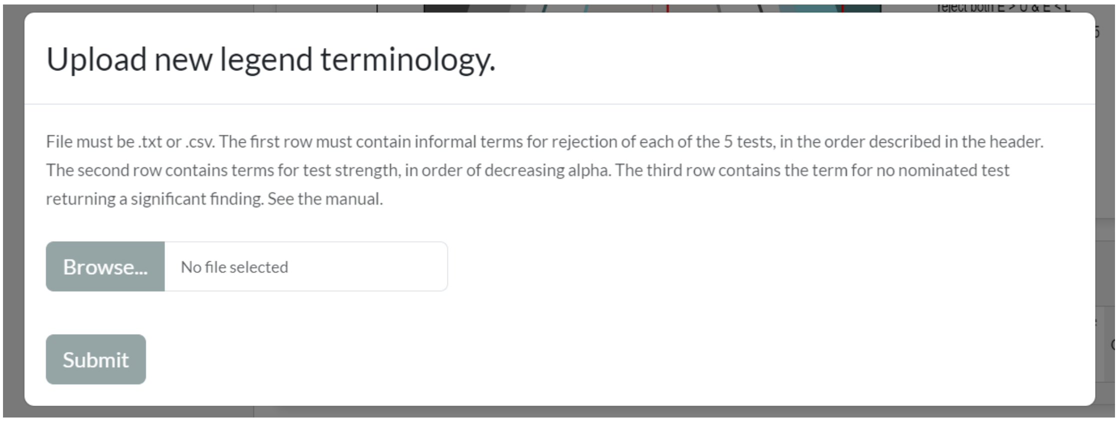
The *Monochrome chart* switch renders the chart in monochrome. Turning it off returns the chart to its default color palette. The monochrome option may be useful when preparing a manuscript.

4.5.2 Changing the color palette

To change some or all the colours in the chart, use the ***Change chart colors*** switch which opens a panel to browse for a color-specification file. Turning the switch off reverts to the default color palette. The color-specification file must be .txt or .csv, with colors identified by name or RGB. The first row contains colors for regions in which the hypothesis E ≥ L is rejected* at each of the alpha levels you are considering (maximum three). The second row contains colors for rejection regions of the hypothesis E ≤ U, while the third row contains the colors for regions in which both E < L and E > U are rejected. This is explained further in the next section. (* In the design phase, regions are pairs of anticipated effect size and total sample size for which the test will have the desired power at the specified alpha level.)

4.5.3 Changing terminology in the legend

An important display option is to change the way that the hypotheses and test levels are reported in the chart legend and in the test table. While some academic journals require the formal descriptions that are the tool’s default, others might prefer clinical



trial terminology, with findings presented as “inferiority”, “superiority” etc. More relaxed terminology can also be very helpful when dealing with stakeholders who aren’t comfortable with formal statistical jargon.

The terminology is changed with the ***Use informal terminology*** switch, which opens the pop-up above. Don’t forget to press the submit button after your selected file has been uploaded. Turning the switch off reverts to the default terminology.

Your file containing the terminology must be .txt or .csv. Its first row must contain your informal or specialist terms for the five findings:

REJECT $E \geq L$
REJECT $E \leq U$
REJECT $E > U$
REJECT $E < L$
REJECT BOTH $E < L$ AND $E > U$.

The second row contains your terms for the strength of tests at each of your alpha levels, in decreasing alpha order. Up to three levels may be specified. The third and final row is how you refer to situations when no test is rejected. For example, this is the clinical trial terminology supplied in file clinical.csv.

```
INFERIOR, SUPERIOR, NON-INFERIOR, NON-SUPERIOR, EQUIVALENT
weak, medium, strong
NO TEST SIGNIFICANT
```

Terminology adopted by the Magnitude-Based Inference approach that was once quite common in some areas of sports science is in the example files MBI.csv and MBIClinical.csv.

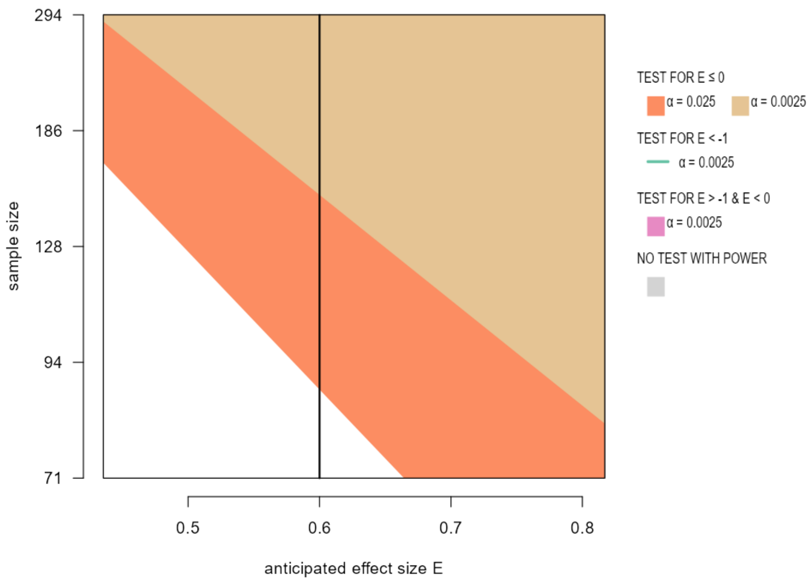
4.5.4 Changing chart ranges

Finally, the ***Change chart ranges*** switch opens a box in which to enter the desired minimum and maximum values. As we said, changing the effect size range also alters the range of the slide bar. The range of the vertical axis is either sample size or standard error, depending on the active tab. Closing the box and turning the switch off will not affect the new ranges.

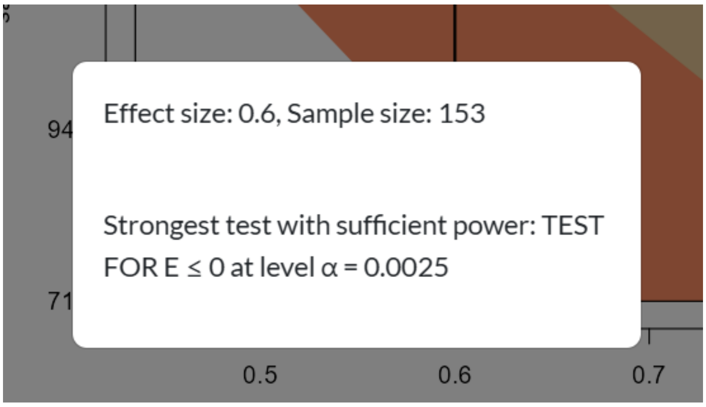
5 USING THE CHARTS

5.1 Calculating sample size

Going back to the Design chart in 3.2, look at the vertical line corresponding to an anticipated effect size 0.6 and where the assumed variance is 1. Given this anticipated effect size and variance, getting 80% power to reject the test $E < 0$ (i.e. testing for superiority) at alpha 0.0025 needs the sample size at which the line crosses into the beige region.



Zooming in on the chart, as shown here, indicates a total sample size of about 150 is required. This estimate can be refined by zooming further, or by clicking on or near the crossing point to get a pop-up like the one below.



A finding of at the weaker level clearly requires a smaller sample size – zooming in again where the line crosses into the orange area gives a total sample size of about 88.

Obviously, tests for inferiority or equivalence aren't going to be any use if the anticipated effect size is greater than the upper margin.

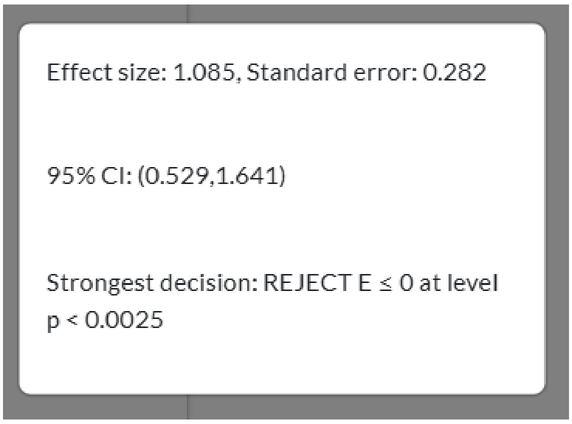
5.2 Interpreting findings

5.2.1 Using the chart in analysis

The vertical axis on the chart is now standard error, which decreases to zero as you go up. As described in 4.4 and 3.1, study summary data, as pairs of effect size and

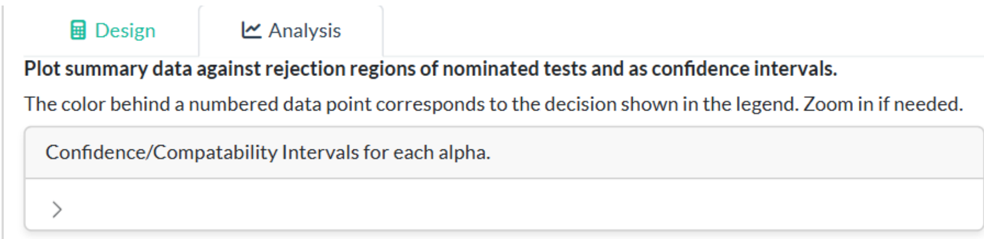
SE or variance, are entered in the side panel and displayed as numbers on the chart.. The region in which the number lies is the strongest finding that can be made. Again, the chart can be zoomed if necessary, and clicking on the chart gives a pop-up with the effect size and standard error at that point as well as the strongest finding.

Sometimes it is useful to look at confidence intervals, also called compatibility intervals (CIs), which are of course ranges of effect sizes. The pop-up opened by clicking on a point on the Analysis chart therefore also gives the 95% confidence interval of a finding with this effect size and standard error.



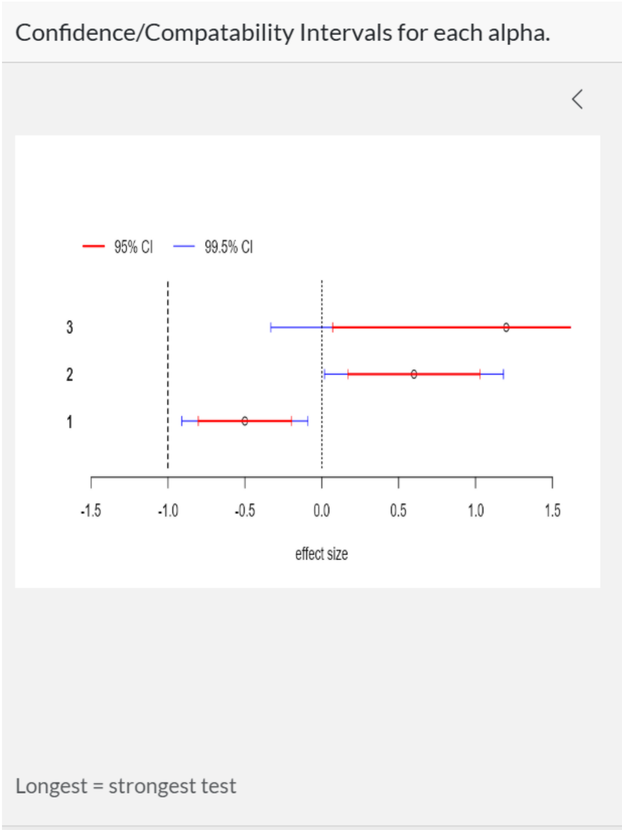
5.2.2 Displaying confidence/compatibility intervals

There is a symmetry between hypothesis tests and Cis which is demonstrated by opening this small pane above the Analysis chart.



With the data points and set up used in 3.1, the opened pane is shown below. The CIs in this screenshot are labelled 1, 2 and 3, matching the numbering of the data points in the chart. Note that a test at a one-sided alpha of α corresponds to a (100 – 2 (1- α))% CI for two-sided CIs.

The CIs look a bit unusual because testing at multiple alpha levels translates to confidence intervals at multiple confidence levels. The higher the confidence, the longer the interval, so here the blue CI corresponding to a 99.5% confidence level is overlaid by the shorter red 95% CI. The vertical dashed/dotted lines are at the margins of the meaningful effect sizes.



The same conclusions can be drawn from the CI display as could be drawn from the chart. For example, for data point 3, the red CI at the 95% confidence level is entirely above the upper margin, allowing a conclusion of superiority at a two-sided test level 0.05. But the longer 99.5% crosses the upper margin, so you can only conclude non-inferiority at this more stringent level.

Opening the CI pane automatically moves the Analysis chart below the depiction of the CIs, but you may have to drag the legend down to align with the chart.

Uncheck the chevron above the CIs to close the pane.